

Vector Infrastructure Cost Reference

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1. Storage assumptions

A single embedding at 384 dimensions stored as float32 occupies 1,536 bytes, approximately 1.5 kilobytes. Metadata adds roughly 0.5 kilobytes per vector, covering the document identifier, chunk identifier, source name, file type, and the chunk text itself. The total per-vector footprint used throughout this reference is therefore approximately 2 kilobytes.

At that footprint, 100,000 vectors occupy about 0.2 gigabytes, one million vectors about 2 gigabytes, and ten million vectors about 20 gigabytes.

Attached block storage on general-purpose SSD is priced at 0.08 dollars per gigabyte-month. Object storage in the standard class is priced at 0.023 dollars per gigabyte-month. The block storage figure is used because the embedded store in this architecture runs on local attached disk rather than against an object-store-backed table.

2. Marginal compute is not zero

The most attackable claim in any embedded-versus-managed cost comparison is that marginal compute is zero. Processor time genuinely is near-free at the assumed volume of 50,000 queries per month, which is 0.019 queries per second; an embedding pass and an index probe at that rate are a rounding error on any host.

Memory is not free. An embedded store's residency cost is memory, and it does not vanish merely because the process is shared with the application. At ten million vectors, quantized index codes wanting to stay page-cache-resident, refine reads into the raw vector column, the embedding model's own resident set of roughly 0.8 to 1.2 gigabytes, and the store's default two-gigabyte read buffer together push the host from a two-gibibyte instance at about 12 dollars per month to an eight to sixteen gibibyte instance at roughly 49 to 98 dollars per month.

The marginal compute row used in this reference is therefore zero dollars at 100,000 vectors, approximately 12 dollars per month at one million, and approximately 35 to 85 dollars per month at ten million.

3. Managed service pricing models

Managed vector database billing is now overwhelmingly usage-based rather than provisioned-pod-based, which makes the honest comparison one about idle cost and minimum floors rather than about per-query price.

The serverless usage-based model charges read units, write units, and storage against a monthly minimum spend of 50 dollars, applied as the maximum of usage and the floor rather than as the floor plus usage. One read unit corresponds to one gigabyte of namespace scanned per query, with a floor of 0.25 read units per query. Neither the requested top-k nor the inclusion of metadata affects query cost.

The resource-hour model bills provisioned memory, processor, and disk rather than queries, at approximately 0.078 dollars per gigabyte-hour, which is about 57 dollars per month per gigabyte of provisioned memory. Published examples put one million 1536-dimensional vectors near 114 dollars per month with quantization enabled, and ten million near 456 dollars.

The activity-unit model bills unit-hours plus tiered storage, with a flexible tier starting near 45 dollars per month and a higher tier near 280 dollars per month on annual terms.

4. Where the comparison actually lands

Against provisioned-capacity services the embedded approach wins by roughly four to ten times at ten million vectors, because the workload never pays for idle memory it did not request.

Against a serverless usage-based service at low query volume the result at ten million vectors is roughly a wash, and below two gigabytes of storage the managed free tier is outright cheaper. The advantage there is operational rather than economic: no vendor relationship, no egress metering, and no data-residency question.

The direction of the sensitivity is counterintuitive and worth stating plainly. Against usage-based pricing the embedded advantage grows with query volume, not shrinks, because an already-provisioned host absorbs additional queries at no incremental charge while usage-based billing does not.

5. Costs the comparison omits

Managed prices bundle compute; a credible embedded row must include it too, and this reference does.

Not bundled, and not free: multi-region replication and automatic failover; backup and point-in-time recovery, at roughly 0.46 to 1.00 dollars per month for a twenty-gigabyte store; index build capacity, where the ten-million-vector clustering training sample alone is about 1.24 gigabytes before the streaming write phase and realistically wants eight to sixteen gibibytes and several hours; and compaction, without which version accumulation grows disk without bound.

Embedding generation at ingest is effectively free on a local model, costing only processor time of roughly two to six hours for ten million chunks. A paid embedding API at ten million chunks of about 125 tokens each would instead be a one-time charge near 25 dollars, which is roughly fifteen months of the storage bill incurred in a single pass. The local choice avoids it entirely, which is worth modelling rather than assuming.